Benchmarking Federated Learning & Knowledge Distillation for Point Cloud Classification

13 federated algorithms $\times$ 10 distillation objectives, 2 datasets, 3 seeds — and an evaluation pitfall

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Two coupled barriers — and one untested pipeline

3D point clouds power robotics, AR and medical imaging, but deploying a classifier where the data actually lives meets two coupled barriers:

- **Privacy**: scans sit in separate clinics or companies and cannot be pooled → train with **federated learning** (FL) [3].
- **Edge compute**: a full **PointNet++** [4] is often too heavy for a portable scanner → compress it with **knowledge distillation** (KD) [1].

The natural pipeline is **FL, then KD**. It is usually judged end to end, by the student's accuracy alone.

The question

Does the compressed student's accuracy **reflect the federated teacher** — or something else entirely?

The gap

To our knowledge, no benchmark evaluates FL and KD *together* for 3D point clouds, let alone on clinical shape data.

The benchmark: a two-stage FL → KD pipeline

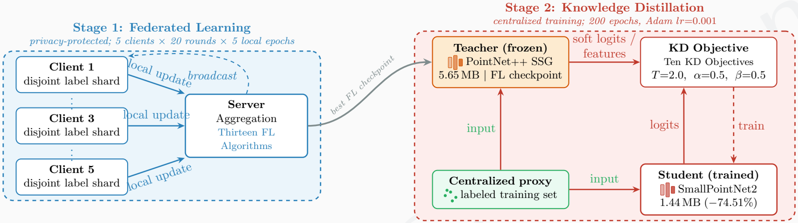

Figure 1: Stage 1 (privacy-protected): **5** non-IID clients + a server running one of **13** FL algorithms train a **PointNet++ SSG** teacher. Stage 2: its best checkpoint supervises a compact student on a *labeled* centralized proxy set with one of **10** KD objectives.

Scale

13 FL × **10** KD → supports all **130** pairs per dataset; **504** standardized runs over **3** seeds.

Data

ModelNet40 [6] (**40** classes); clinical **craniosynostosis** head shapes [5] (**4** classes). **Worst-case label skew**: disjoint label slices per client.

Models (on ModelNet40)

Teacher **PointNet++ SSG**, **5.65 MB** → student **SmallPointNet2**, **1.44 MB**. **20** rounds × **5** local epochs.

Finding 1: federated training degrades sharply under label skew

Instance acc. (%) ↑	ModelNet40	Clinical
Centralized reference	92.26	100.00
Best FL method	76.32 FedNova	75.83 FedProx
Gap to centralized	−15.94	−24.17
4 server-side optimizers	4.05–6.28	25.00–26.25
chance level	2.5	25

Best checkpoint per run, mean over 3 seeds.

No universally robust algorithm

The ranking is **dataset-dependent**: **FedNova** leads on **ModelNet40** but is mid-pack on the clinical data, where **FedProx** leads. **FedAvgM**, **FedAdam**, **FedYogi** and **FedAdagrad** collapse to near-chance on both.

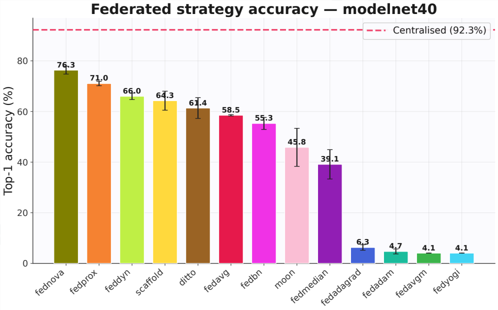

Figure 2: Standalone FL accuracy on **ModelNet40** (mean $\pm$ std, 3 seeds) against the centralized reference.

Finding 2: distillation compresses the teacher at near-ceiling accuracy

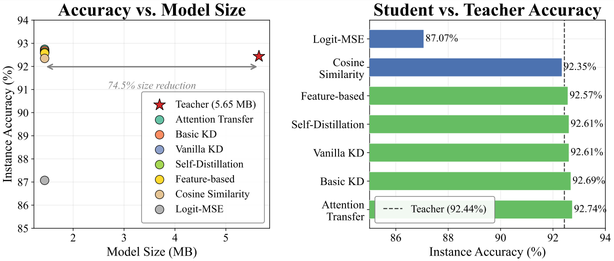

Figure 3: **ModelNet40**, focused round: instance accuracy vs. model size (left); each student vs. the **92.44%** teacher (right). Six of the ten objectives, plus a full-weight **Basic KD** probe.

✔ **74.51%** smaller: **5.65 MB** → **1.44 MB**

✔ ≈**2**× faster inference (≈**50%** lower latency)

✔ **5** of **7** objectives match or beat the teacher (**92.57–92.74%**)

✘ **Cosine** trails by **0.09**; **Logit-MSE** is the real outlier, **87.07%**

💡 With a strong teacher, almost every objective succeeds — so standalone KD barely tests it.

Finding 3: a hard-label term masks a collapsed teacher

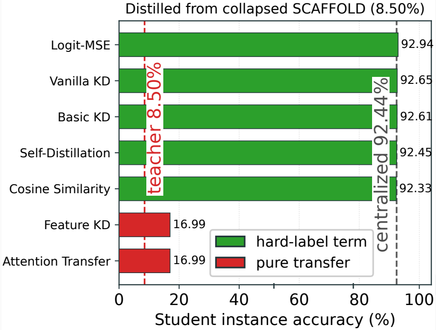

Figure 4: **ModelNet40** focused round: students distilled from a collapsed **SCAFFOLD** checkpoint — **8.50%** standalone here, vs. its **64.26%** multi-seed mean.

Every objective shares one loss

$$\mathcal{L} = w_{\mathrm{CE}}\,\mathcal{L}_{\mathrm{CE}}(\hat{y}, y) + \mathcal{L}_{\mathrm{transfer}}$$

w_{CE} weights the hard-label cross-entropy on the *labeled proxy set*.

- $w_{\mathrm{CE}} > 0$ (**Vanilla KD**, **Basic KD**, **Self-Distillation**, **Logit-MSE**, **Cosine**): the student learns the proxy labels. **SCAFFOLD** teacher at **8.50%** + **Logit-MSE** → **92.94% student**: an **84.4**-point gap that no transfer could explain.
- $w_{\mathrm{CE}} = 0$ (**Attention Transfer**, **Feature KD**): only the teacher remains → **16.99%**, tracking it.

The illusion survives on real clinical data

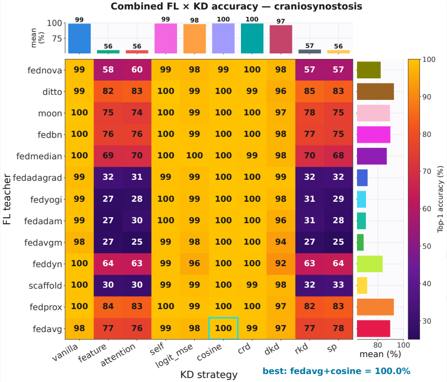

Figure 5: Full **13**×**10** clinical grid, all **10** objectives (mean of **3** seeds). Read the *pattern*: bright columns keep a hard-label term, dark ones do not.

6 objectives with a hard-label term

Recover from *any* teacher: spread $\leq$ **7.5** points across **13** teachers, r from $-$**0.53** to $+$**0.37**. A chance-level **FedAvgM** teacher (**25%**) still yields **97.5%** (**Vanilla KD**), **100%** (**CRD**).

4 label-free objectives

Feature KD, **Attention Transfer**, **RKD**, **SP** track it: $r =$ **0.986–0.990**. Behind the chance teacher they give **25.0–27.5%**; behind **FedProx** (**75.83%**), **81.7–84.2%**.

⚠ The cross-entropy term reuses the very patient labels that federation was meant to protect.

Takeaways

1. ★ **Label skew breaks federated 3D training** — best method **15.94** (**ModelNet40**) and **24.17** points (clinical) below centralized; the ranking flips across datasets.
2. ★ **Distillation compresses well** — **74.51%** smaller, $\approx$**2**$\times$ faster, near-ceiling accuracy.
3. ★ **Hard-label distillation measures the proxy labels, not the federated teacher** — a collapsed teacher at **8.50%** still "yields" a **92.94%** student; label-free objectives track the teacher ($r \approx$ **0.99** on the clinical grid).

Recommendation

Evaluate FL–KD pipelines with label-free distillation: drop the cross-entropy term ($\alpha=1$) or distill on an *unlabeled* proxy, e.g. by aggregating client soft predictions [2], so the reported accuracy reflects the federated teacher.

Benchmark, code and every checkpoint: `ezharjan.github.io/FLKD3DBenchmark`

Thank you!

Questions?

`ezharjan.github.io/FLKD3DBenchmark` | **ezharjan@outlook.com**

References

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